An Implementation of Open-RMF in an office environment

Objects

- Build 2 AMR
  - Mini from scratch
  - Jobot port existing code from ROS 1 to ROS 2
- Port the DStar global planner
- Build an infrastructure to control this robot using Open-RMF

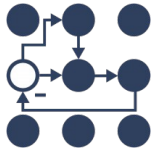
Build 2 AMR

- Why ROS?
- Why ROS 1

EOL

ROS 2

Open-RMF

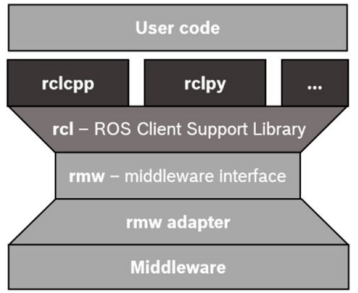


ROS

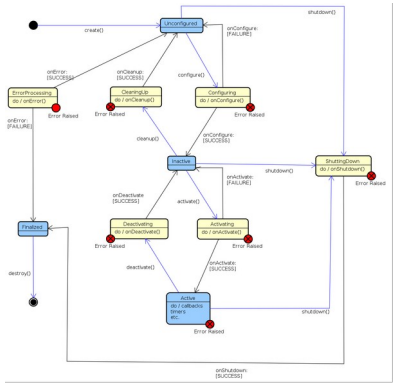
MoveIt

ROS Components

- Middleware
  - DDS (eProsima Fast DDS, Eclipse Cyclone DDS, …)
  - No DDS (Eclipse Zenoh)
- Code:
  - Multiple client libraries : rclc, rclcpp, rclpy, rclrs, …
  - Topics, Service, Action
  - Lifecycle
  - Composition



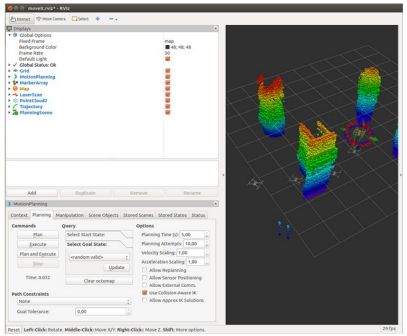
Structure



Lifecycle

ROS Components

- Tools
  - Launch file (python, xml, ...)
  - Rqt
  - Visualization Rviz
- External tools
  - Gazebo

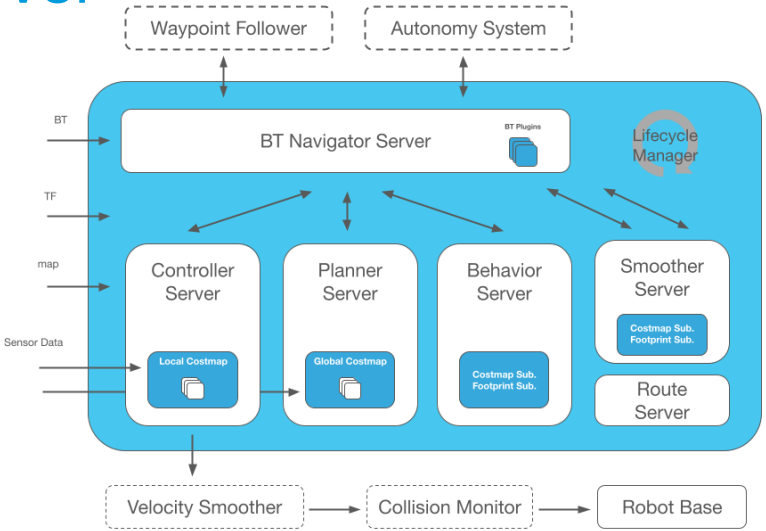


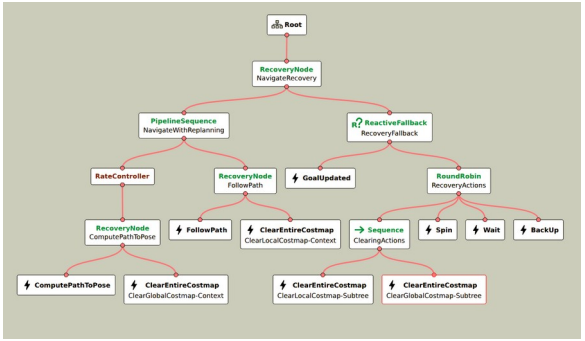
Rviz

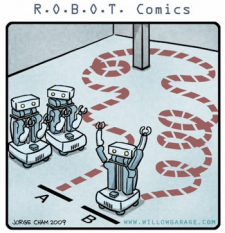
Gazebo

Nav2 project

- BT Navigation Server
- Behavior Server
- Planner Server
- Controller Server
- Smoother Server
- Route Server
- Lifecycle manager







Nav2 Requirements

- Need a TF tree based on REP 105
  - base_link → odom → map
    - base_link → odom (continuous link)
      - Given by robot estimation (wheel encoders, …)
      - ROS Packages (robot_localization, fuse, ...)
    - Odom → map (can be discrete link)
      - SLAM (Cartographer, …)
      - AMCL
      -

Build Mini AMR

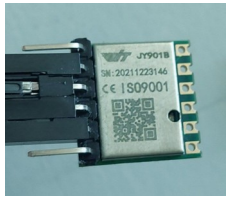
Unmanned Ground Vehicle

Control movement and give information about its state

Sensors

Laser Scan

IMU

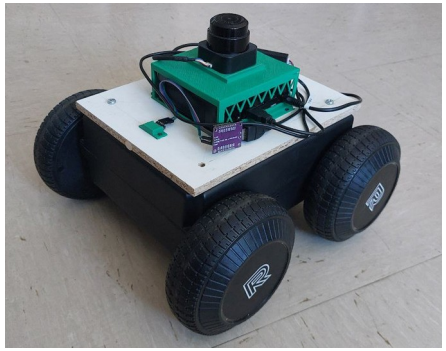


Give information about the world or give more data

PC

Take all the data and control the robot without human interaction

AMR



Upgrade Jobot AMR

With serial communication through arduino on PC with ROS1

IMU
Laser

Upgrade Jobot AMR - ROS 1→ ROS 2

- ROS msgs update
- Find ros-serial for ROS 2
  -
- Lifecycle implemented
  - In configure() add creation of secondary nodes

Updated DStar global planner and results

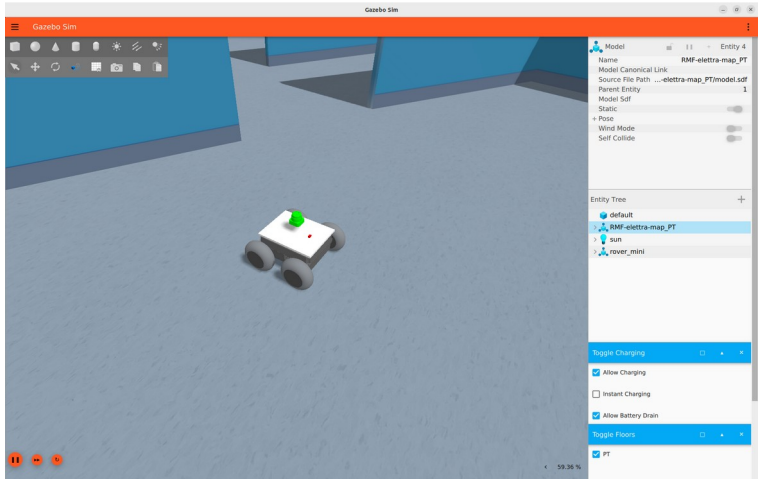
- Upgrade to ROS 2
  - Minor changes, similar structure to ROS 1

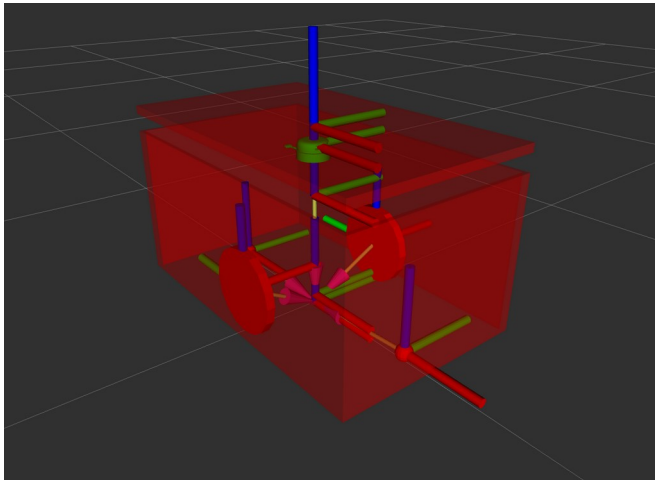
- Results:

Start->End	ROS 2 D-start (s)	ROS 2 NavfnPlanner (s)
office -> storage room	1.00	0.020
office -> entrance	5.47	0.150
entrance -> CAD designers	5.36	0.130

Adding also simulation

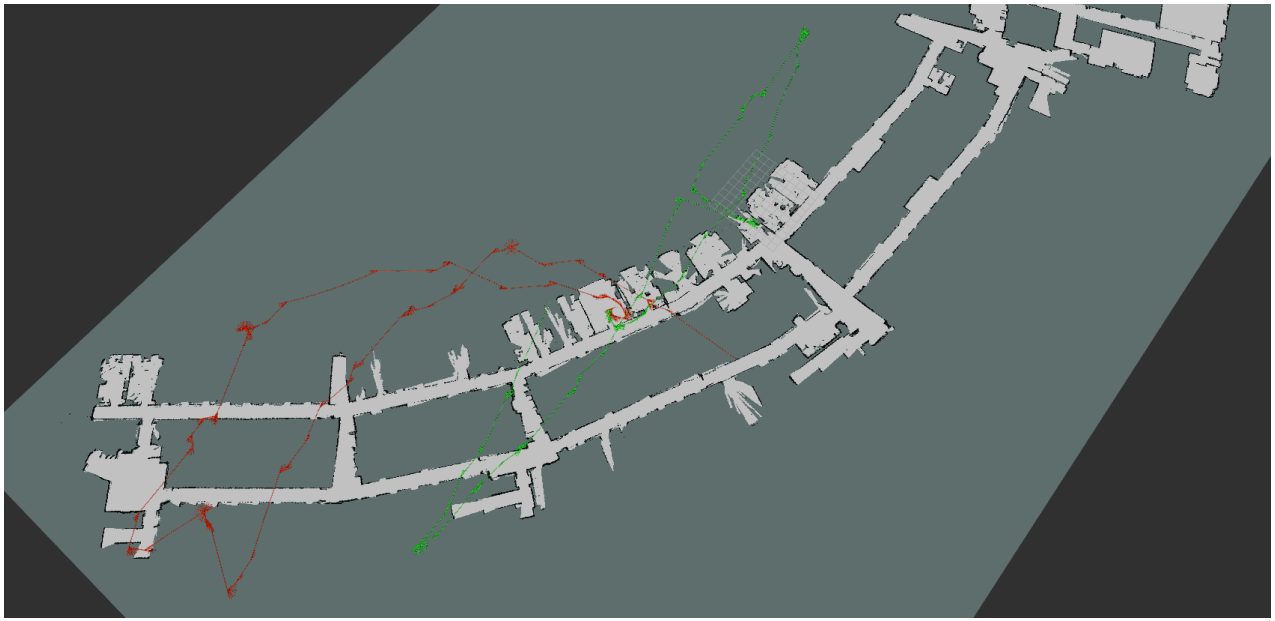
- Different drive 4WD Mini and 2WD Jobot
- TF tree

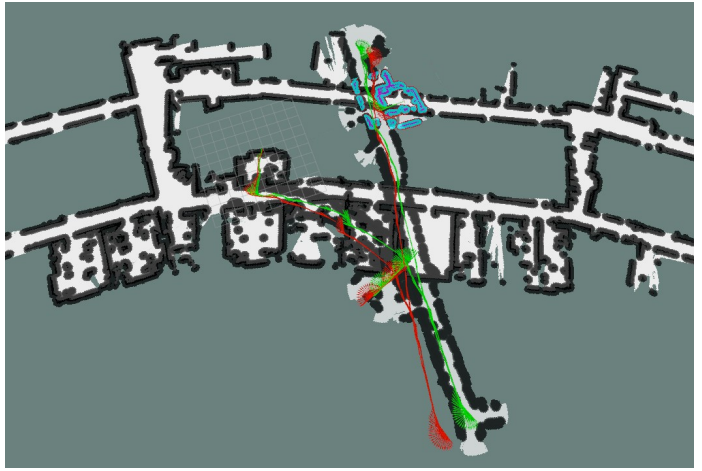




Move Base - Problems

- Their returned odometry → need to improve





How to improve Odometry

- Using packages that fuse sensors data:

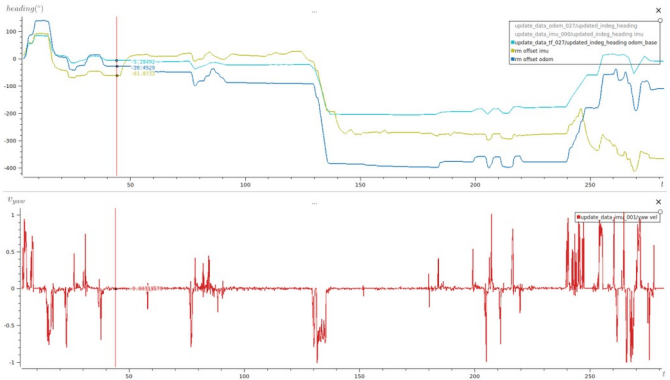
  robot_localization that fuse IMU data with odometry given by robot
- And tuning

  tuning the parameters of the EKF

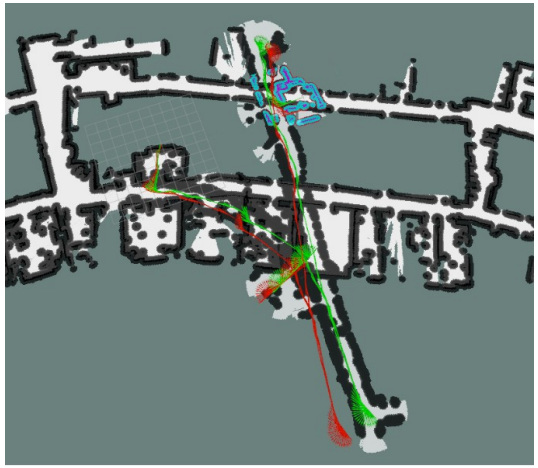
- Moving the robot and recording the data using rosbag2
  - Repeatability
  - Speed

How to tune EKF of robot_localization

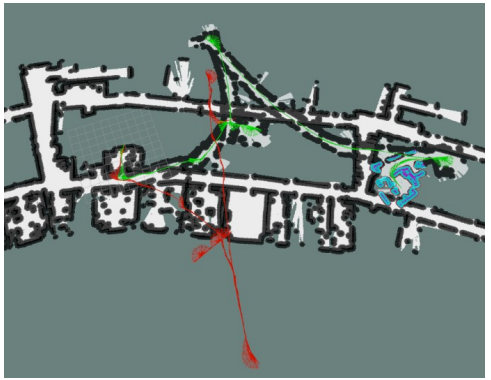
- Choose what sensors use and how



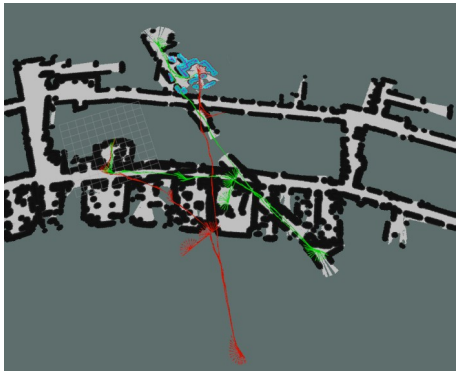
Mini error in heading



Jobot with yaw from odometry



Jobot with v_{yaw} from odometry



Jobot with v_{yaw} from odometry and IMU

How to tune EKF of robot_localization

- Change the process noise covariance matrix Q

$$x_k = f(x_{k-1}) + w_{k-1}$$
$$z_k = h(x_k) + v_k$$

Prediction step

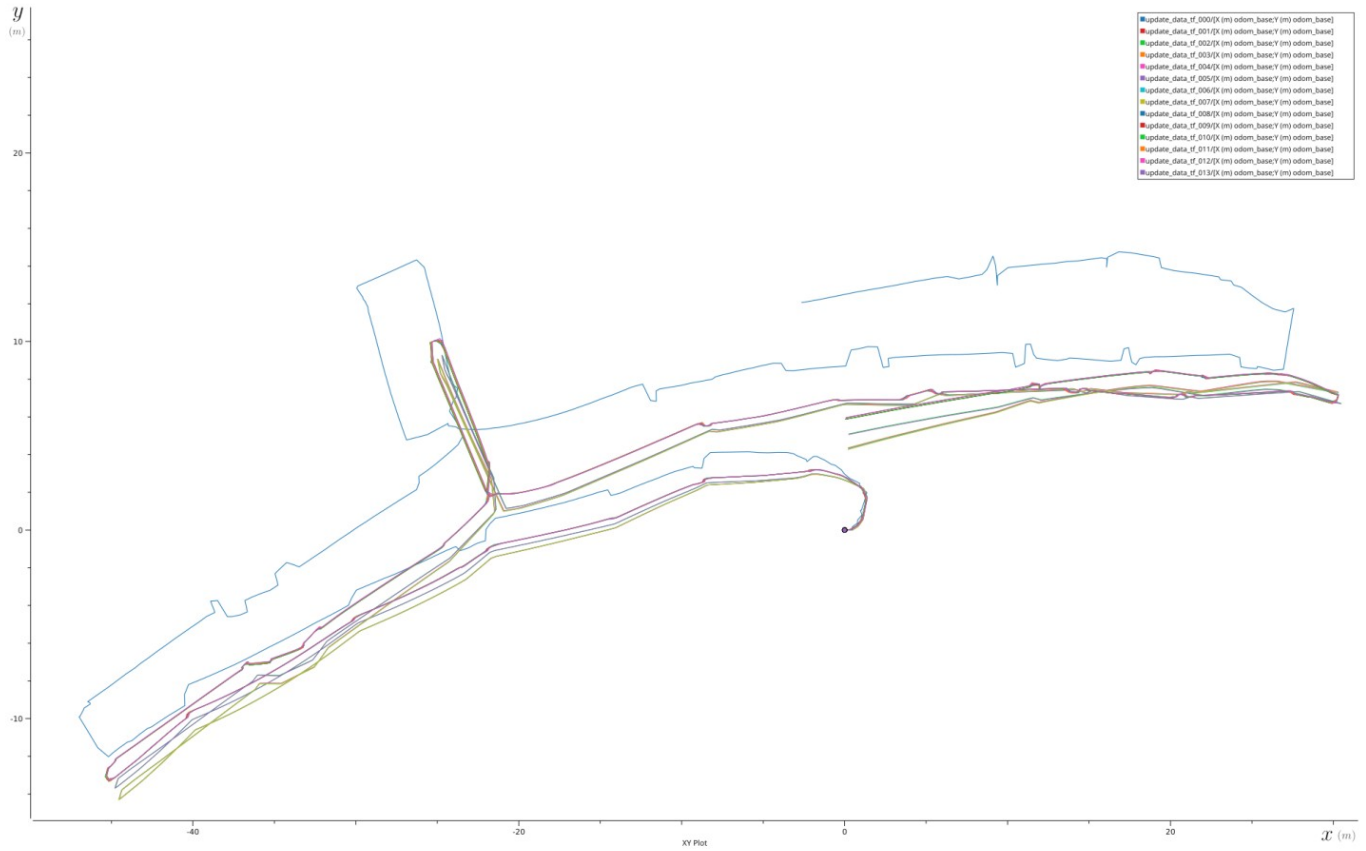
$$\hat{x}_k = f(x_{k-1})$$
$$\hat{P}_k = F P_{k-1} F^T + Q$$

correction step

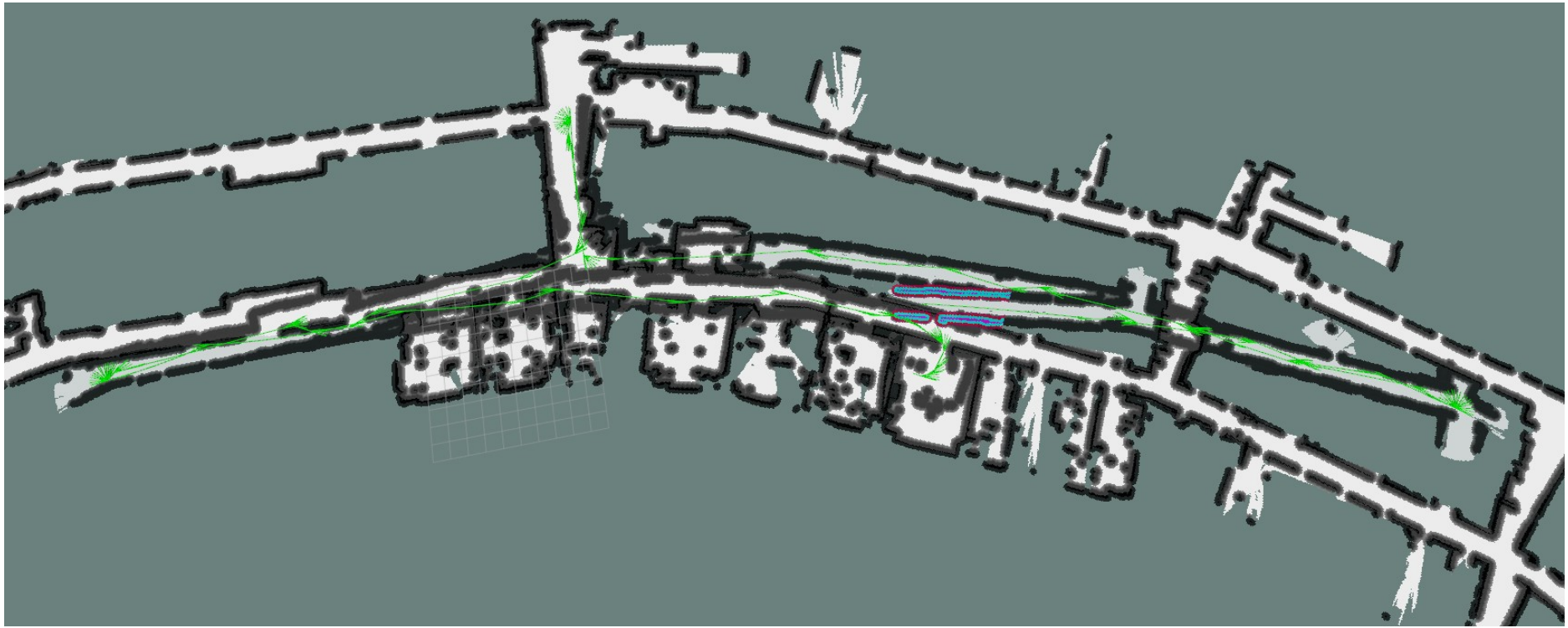
$$K = \hat{P}_k H^T (H \hat{P}_k H^T + R)^{-1}$$
$$x_k = \hat{x}_k + K(z - H\hat{x}_k)$$
$$P_k = (I - KH)\hat{P}_k (I - KH)^T + KRK^T$$

Results Mini

v_x (-)	v_y (-)	v_{yaw} (-)	data name	diff start end (m)
0.001	0.001	0.001	000	12.36
0.01	0.01	0.01	001	5.88
0.1	0.1	0.1	002	5.10
10.0	10.0	10.0	003	4.37
50.0	50.0	50.0	004	4.31
0.001	0.001	0.1	005	4.83
0.001	0.001	10.0	006	4.33
0.001	0.001	50.0	007	4.27
0.001	0.1	0.001	008	5.88
0.001	10.0	0.001	009	5.88
0.001	50.0	0.001	010	5.88
0.1	0.001	0.001	011	5.94
10.0	0.001	0.001	012	5.94
50.0	0.001	0.001	013	5.97

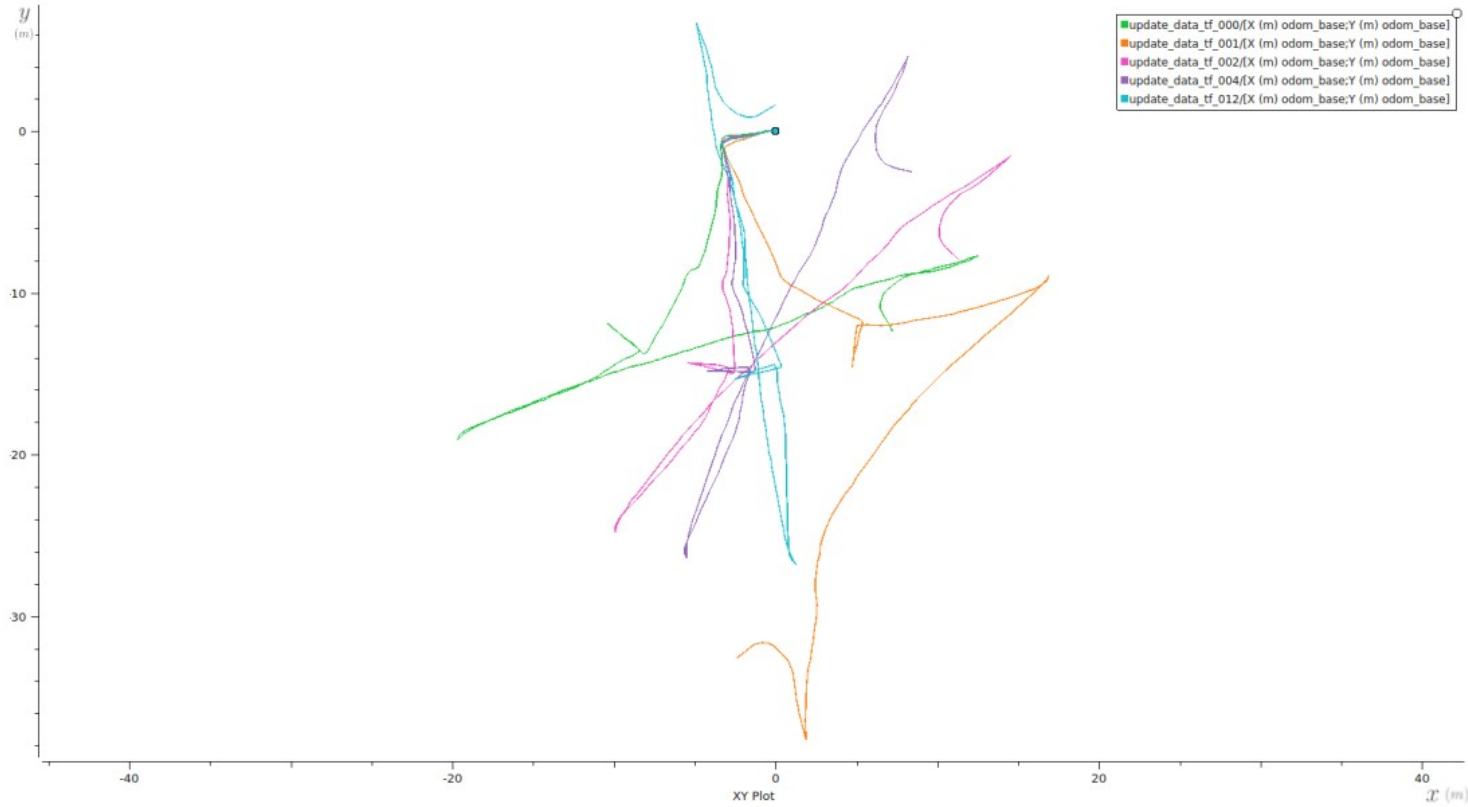


Best result Mini

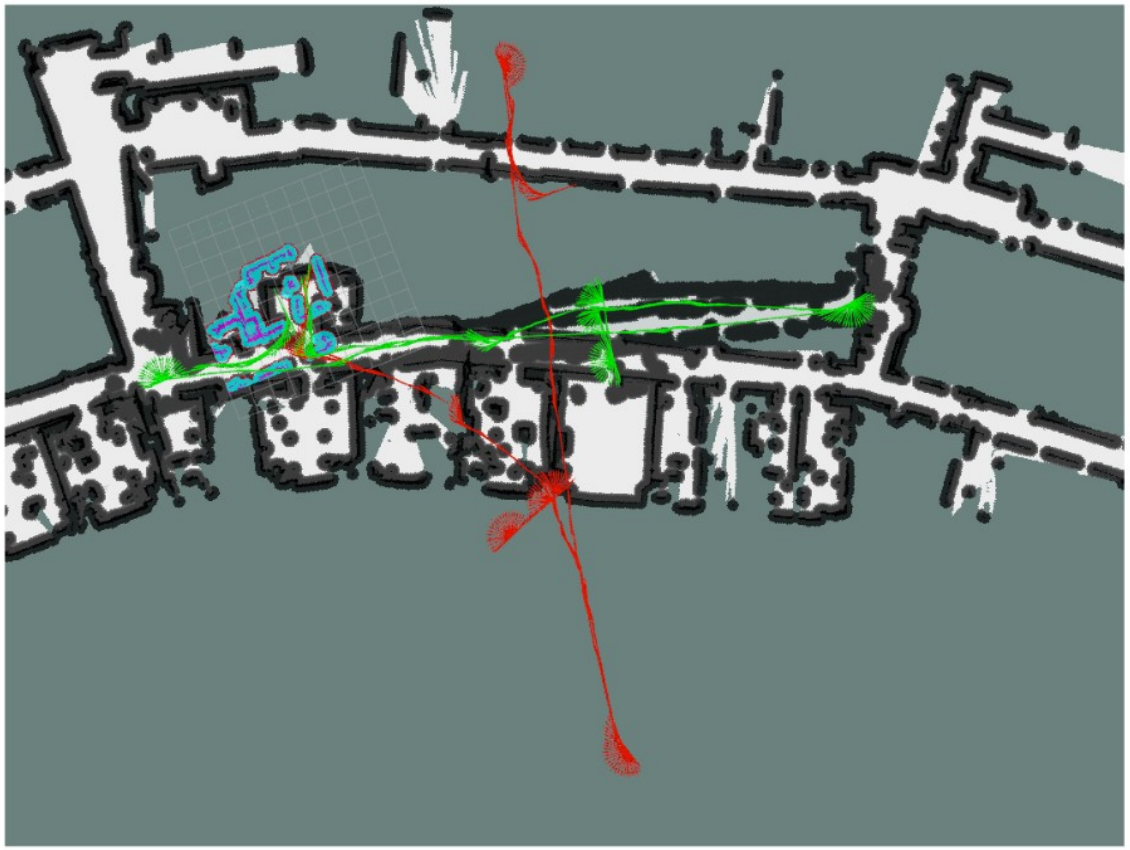


Results Jobot

v_{yaw} (-)	record name	diff start end (m)
-	000	14.28
-	001	32.67
0.1	002	13.88
10.0	003	13.26
100.0	004	8.74
200.0	007	4.31
300.0	008	1.83
400.0	009	3.50
320.0	012	1.63



Best result Jobot

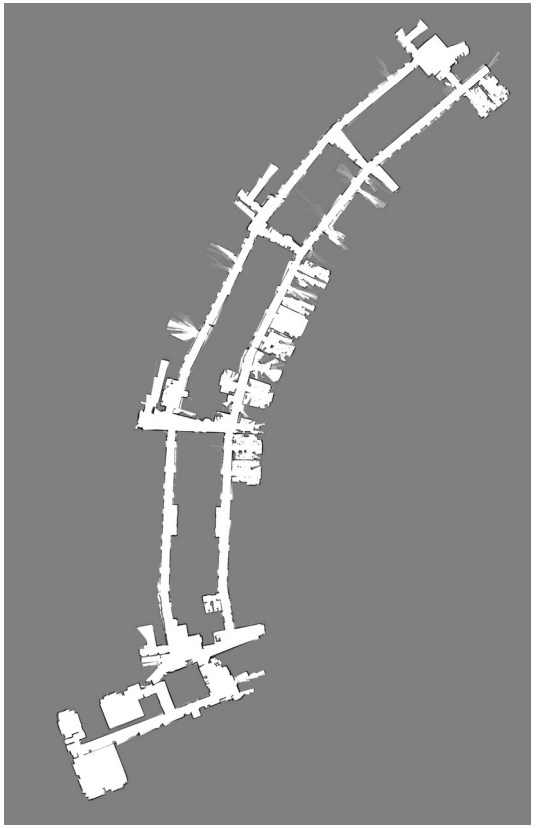


Global localization

We improve but still errors

correction using a map

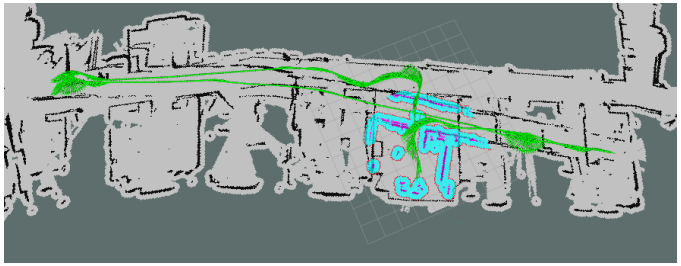
- SLAM or only localization?
  - Localization:
    - Cartographer
    - AMCL



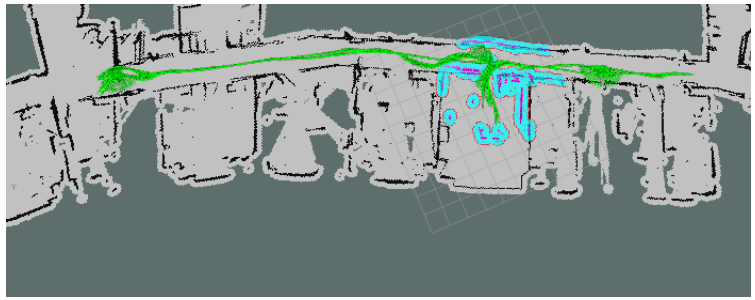
How to tune AMCL

- Tune the expected process noise
  - α1 : Expected process noise in odometry's rotation estimate from rotation.
  - α2 : Expected process noise in odometry's rotation estimate from translation.
  - α3 : Expected process noise in odometry's translation estimate from translation.
  - α4 : Expected process noise in odometry's translation estimate from rotation.

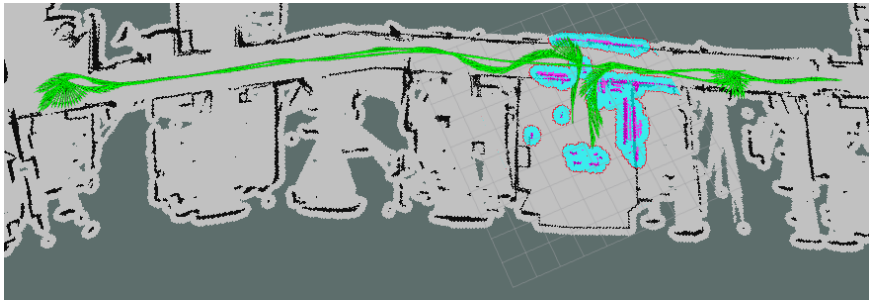
AMCL calibration Mini



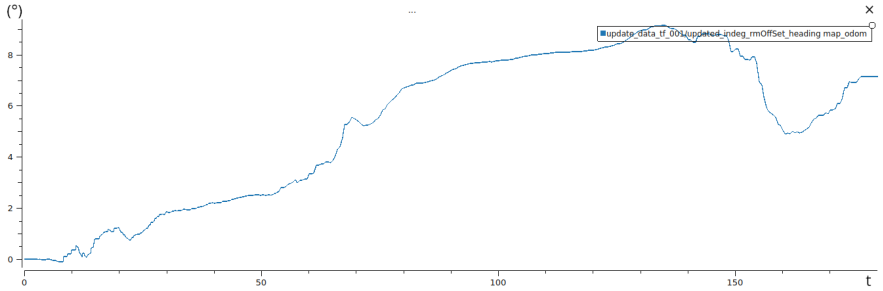
Correcting rotation



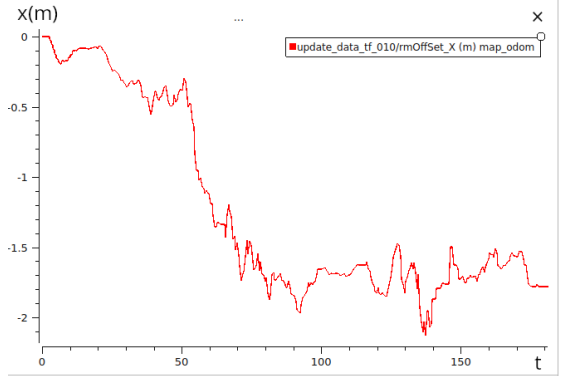
Correcting translation

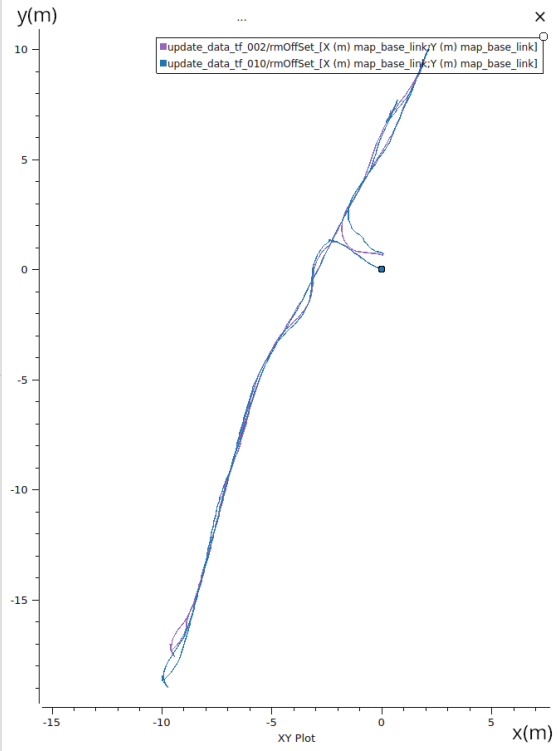


Results Mini correction with AMCL

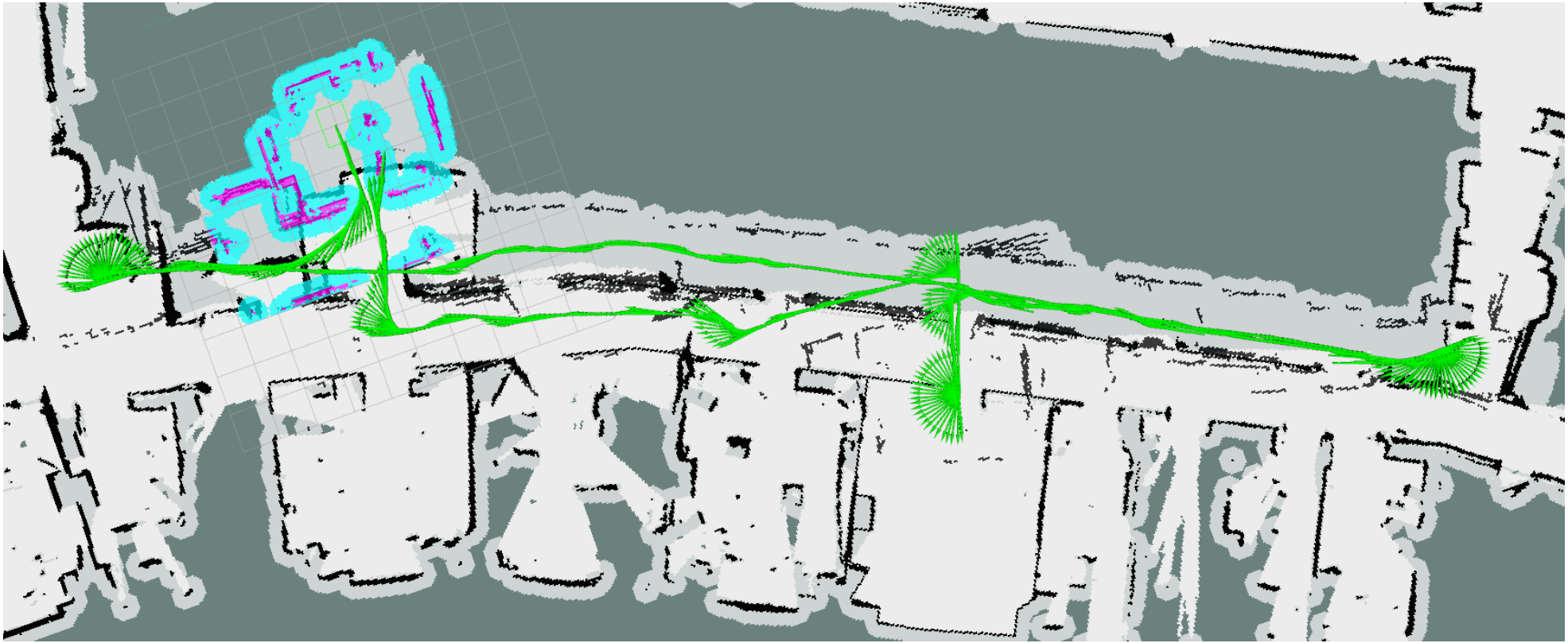


$\alpha 1$ (-)	$\alpha 2$ (-)	$\alpha 3$ (-)	$\alpha 4$ (-)	distance (m)	data name
0.0	0.0	0.0	0.0	1.92	-
0.001	0.0	0.0	0.0	1.32	000
0.01	0.0	0.0	0.0	0.60	001
0.1	0.0	0.0	0.0	0.64	002
0.0	0.001	0.0	0.0	0.58	003
0.0	0.1	0.0	0.0	0.51	004
0.0	0.0	0.001	0.0	2.01	005
0.0	0.0	0.01	0.0	2.27	006
0.0	0.0	0.1	0.0	2.32	007
0.0	0.0	0.0	0.1	3.08	008
0.01	0.01	0.01	0.0001	0.53	009
0.01	0.01	0.1	0.0001	0.69	010



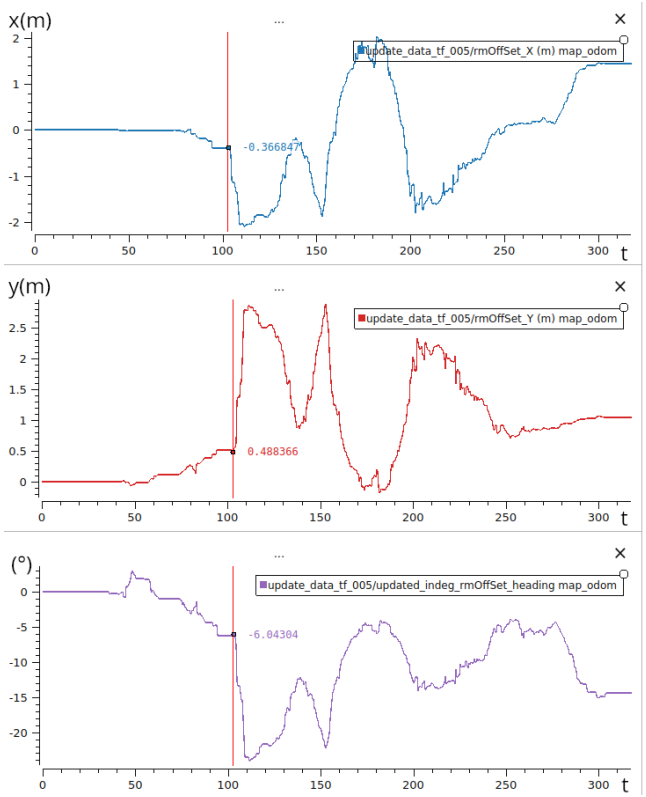


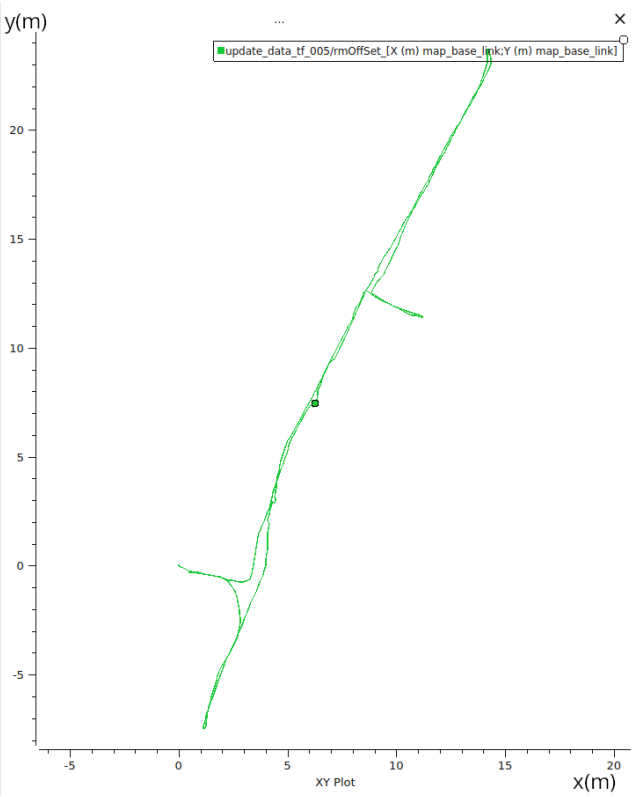
First correction Jobot



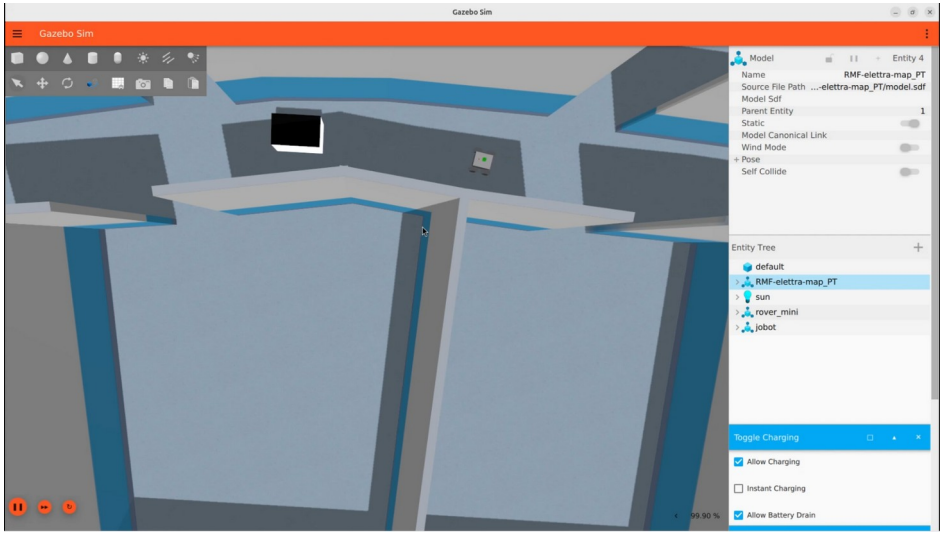
Results Jobot

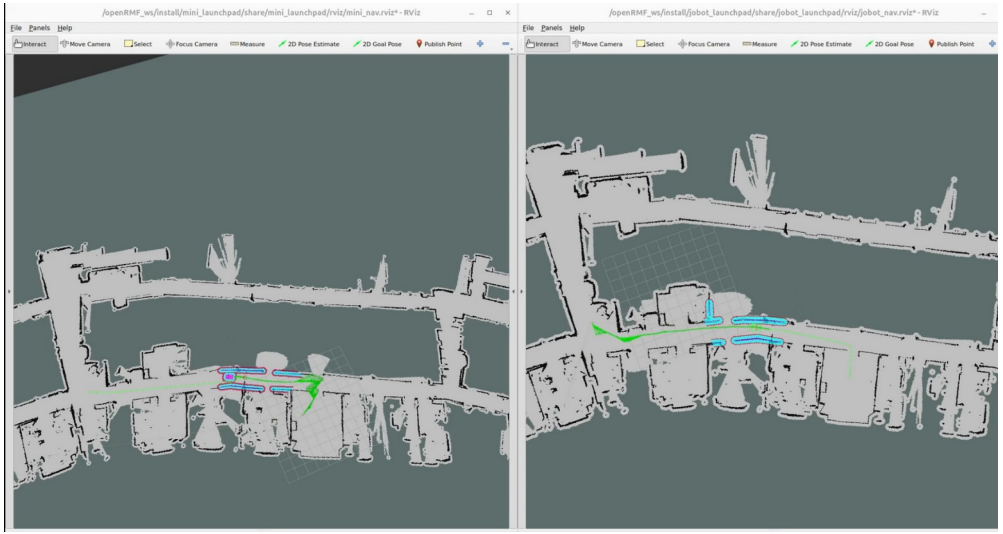
$\alpha 1$ (-)	$\alpha 2$ (-)	$\alpha 3$ (-)	$\alpha 4$ (-)	distance (m)	data name
0.0	0.0	0.0	0.0	1.63	-
0.001	0.0001	0.0001	0.0001	1.58	000
0.01	0.0001	0.0001	0.0001	2.16	001
0.01	0.001	0.0001	0.0001	0.71	002
0.01	0.001	0.001	0.0001	0.87	003
0.01	0.001	0.001	0.0001	0.74	004
0.01	0.001	0.001	0.001	0.59	005





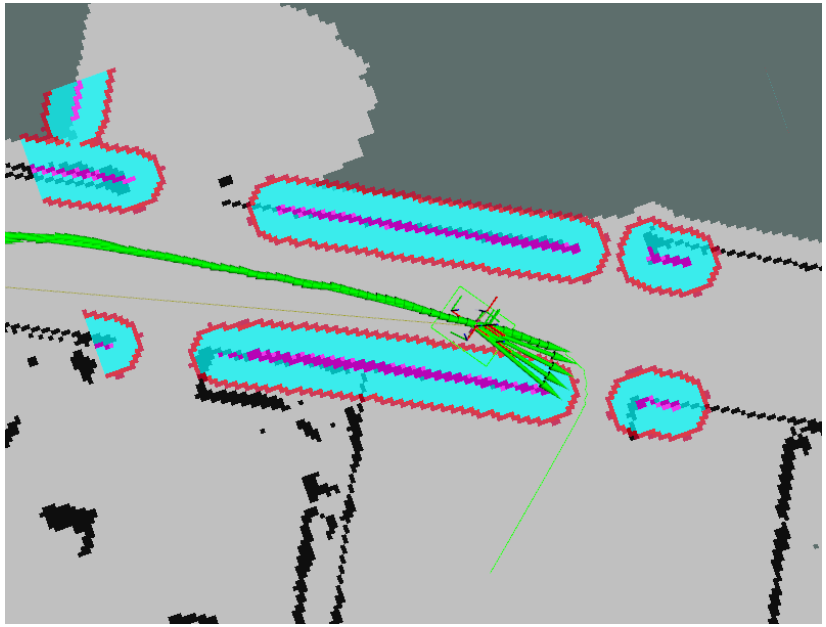
Results in simulation





Discovery in simulation

Error with the controller

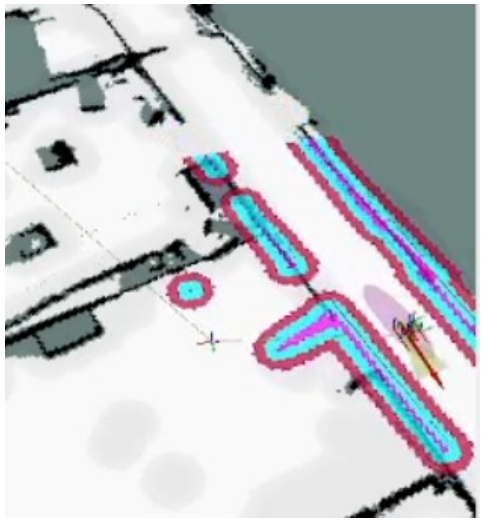


DWB Controller

MPPI Controller

Discovery in reality

- Miss alignment from local costmap and global costmap



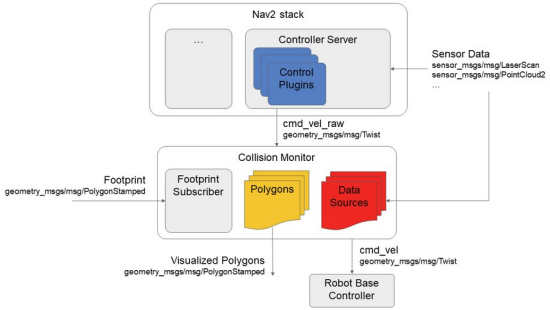
System fail to generate
a path to enter the room

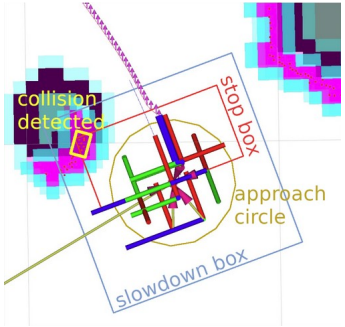
System trying to turn left before recover
thanks to the recovery action in BT

Discovery in reality

- Obstacle detection not visible

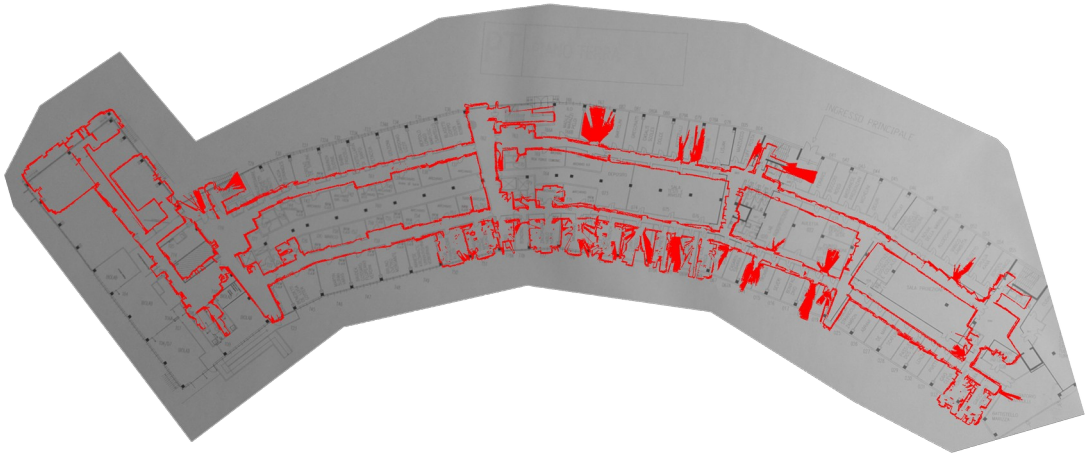
- Improve safety





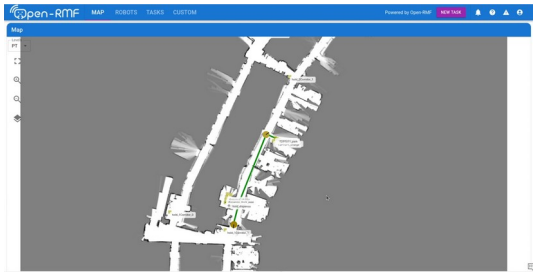
Discovery in reality

- Misalignment Cartographer and building map

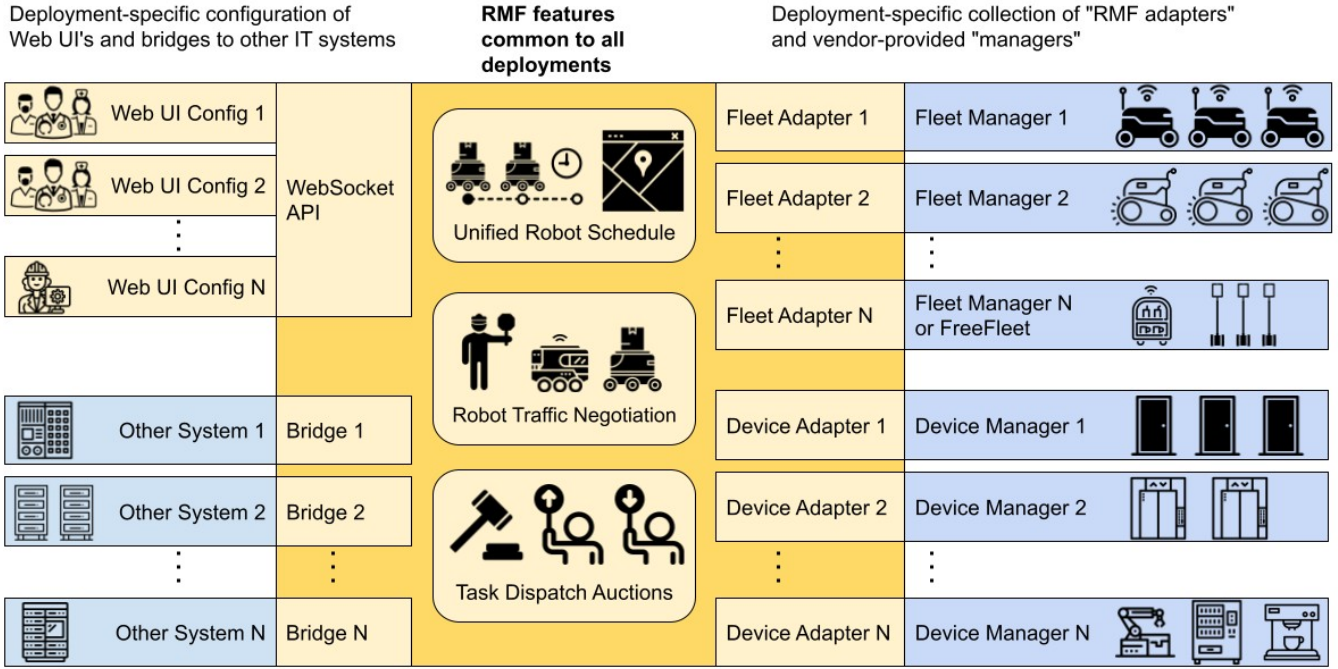


- Communication
  - ROS_DOMAIN_ID
  - DDS properties
  - namespace

Build a system manager for the fleet Mini and for the fleet Jobot

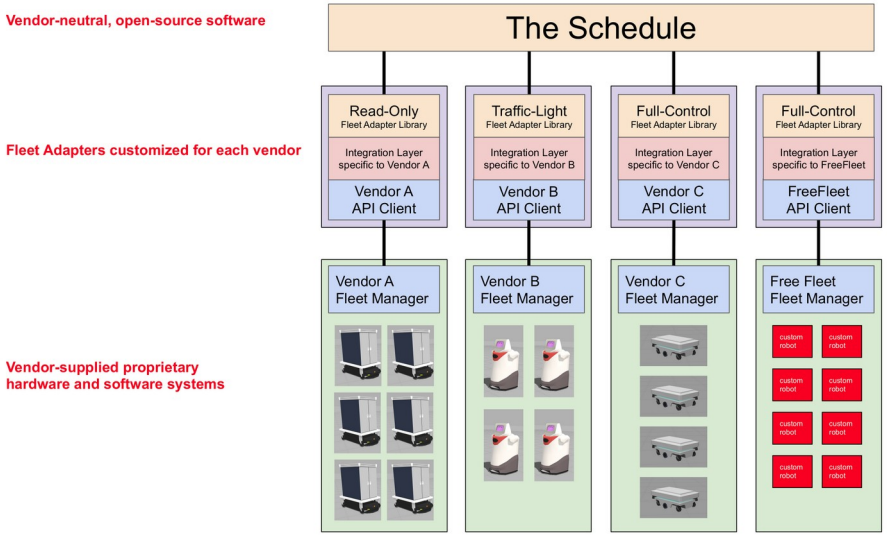


rmf_web

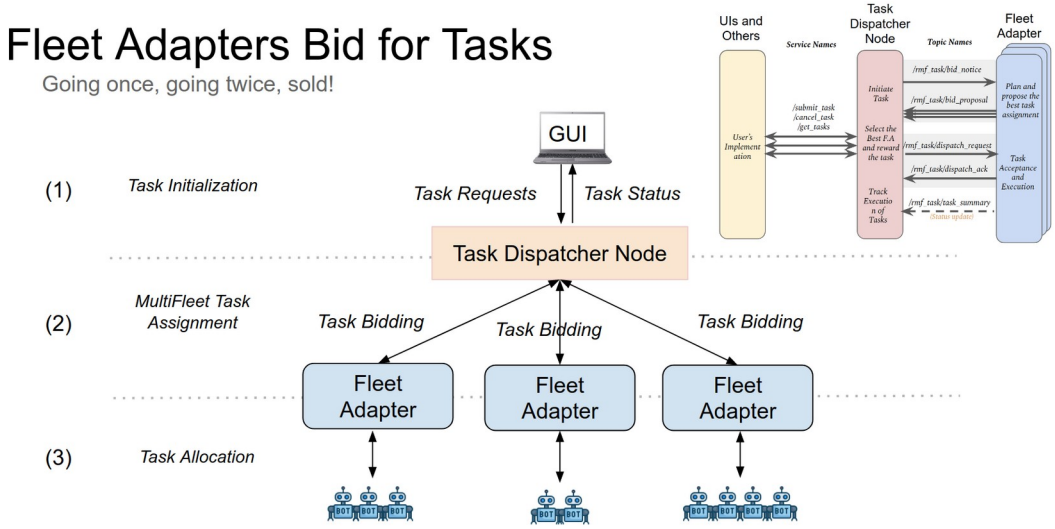


Open-RMF - Schedule and Bidding

rmf_traffic_schedule

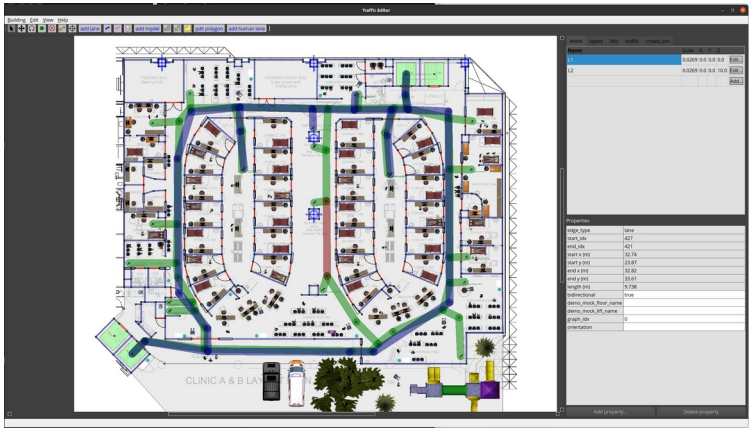


rmf_task_dispatcher



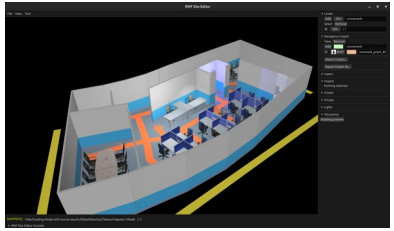
Open-RMF - Navigation graph

rmf_traffic_editor



NEW!

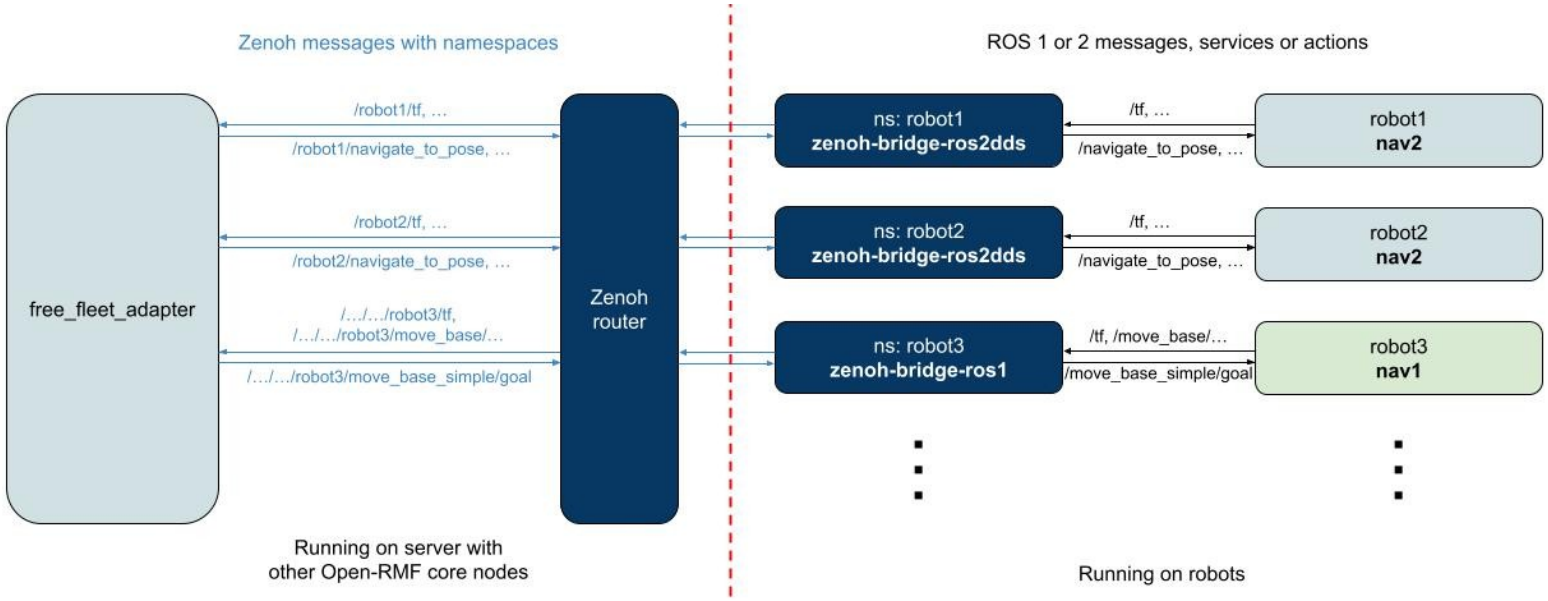
rmf_site



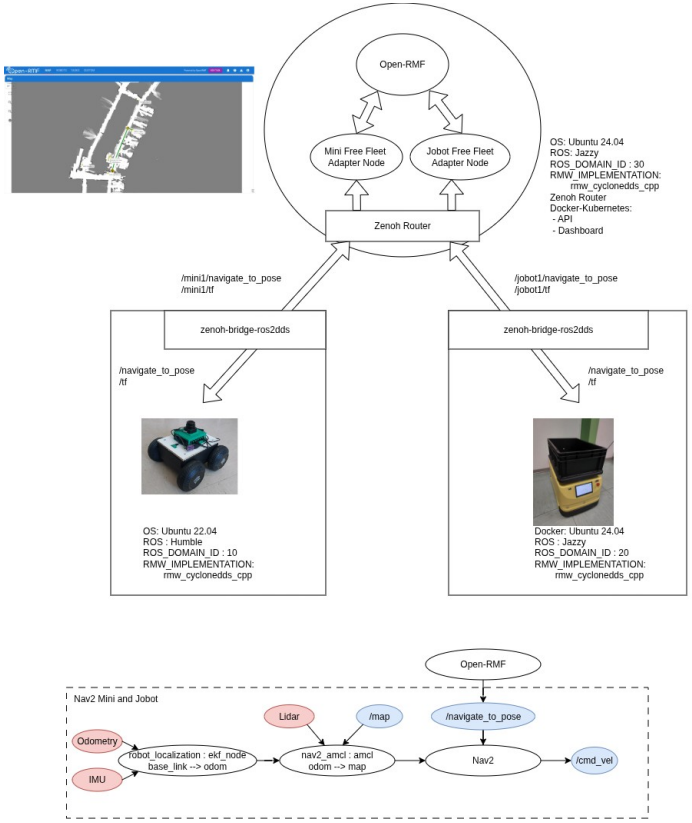
Free Fleet

Free Fleet

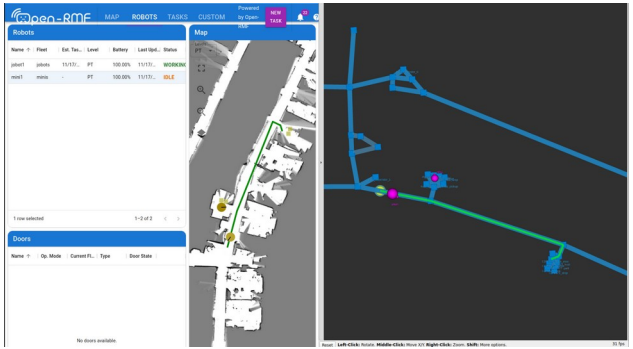
a ROS 2 dedicated Fleet adapter based on fleet_adapter_template



Complete Open-RMF structure

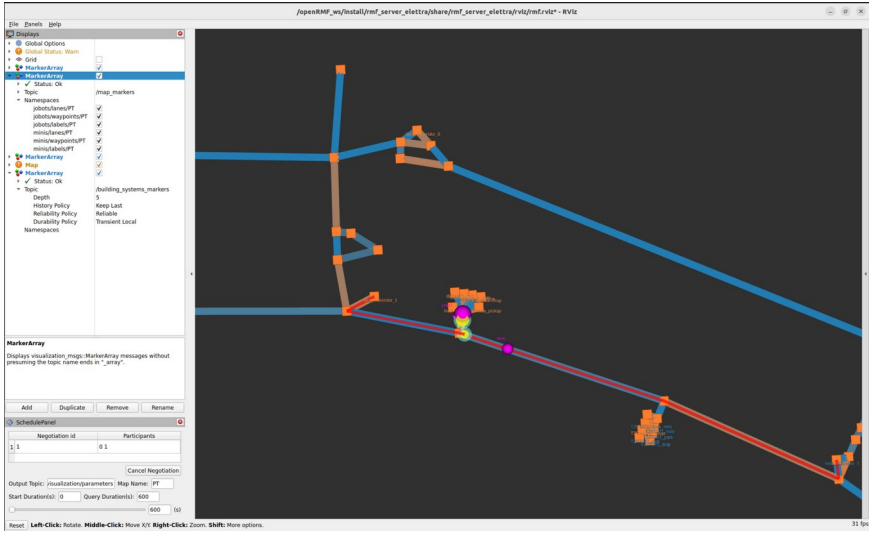


Open-RMF results



Patrol task without negotiation

Mini waiting the Jobot



Patrol task with negotation